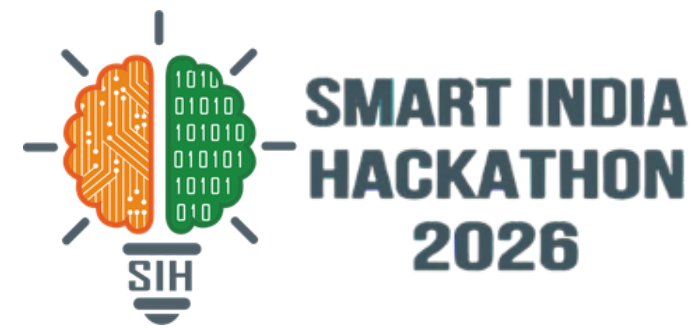
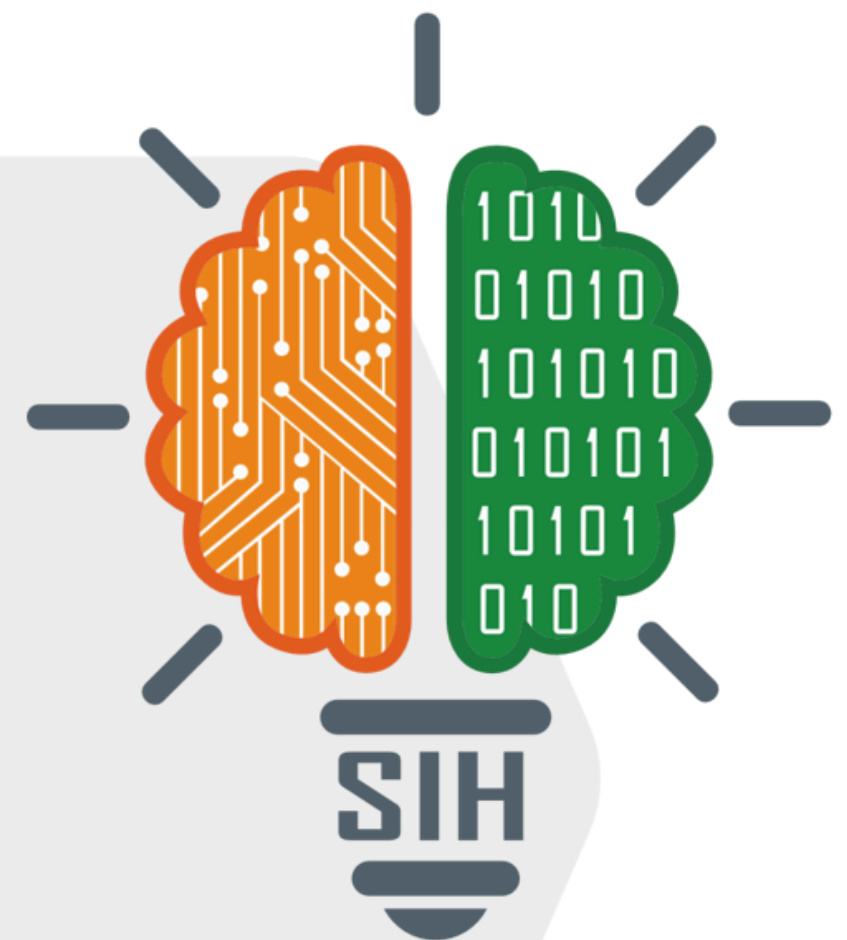


SMART INDIA HACKATHON 2026



- **Problem Statement ID:** SIH26076
- **Problem Statement Title:** Development of personalized homepage for 'Mausam' mobile application
- **Nodal Ministry / Organization:** Ministry of Earth Sciences (MoES) / IMD
- **Theme & Category:** Smart Automation | Software
- **Team Name:** Team Mausam
- **Package:** com.teammausam.mausam_app
- **Github Repo Link:** github.com/Shreyansh303/team_mausam_sih_2026
- **Demo:** https://www.youtube.com/watch?v=dAnlxq_2jj8
- **Target OS:** Android 7.0+ (minSdk 24), ios & Flutter Web



PROPOSED SOLUTION

- Users choose 1–3 personas, and the engine ranks 33 card types by persona affinity to build a personalized feed.
- The feed is easy to understand and adapts over time. Each card shows up to 4 reasons for why it appears, while learning stays limited so important warnings are never buried.
- Urgency comes first, regardless of preference. A warning with urgency ≥ 0.8 stays pinned for all 8 personas, ensuring critical alerts reach everyone.
- Time, season, location, elevation, and active warnings shape what users see. For example, a parent sees School Run first in the morning, but not at night.

WHY WE STAND OUT



Server-driven

Change the homepage from the backend—no app update needed.



Honest data:

Estimated data is clearly marked, with no fake IMD data.



Live updates

New warnings can instantly change what users see.



Multilingual

Translates cards, advice, and actions—not just buttons.



Smart ranking

Cards are ranked based on relevance, context, and urgency.



Made for real phones

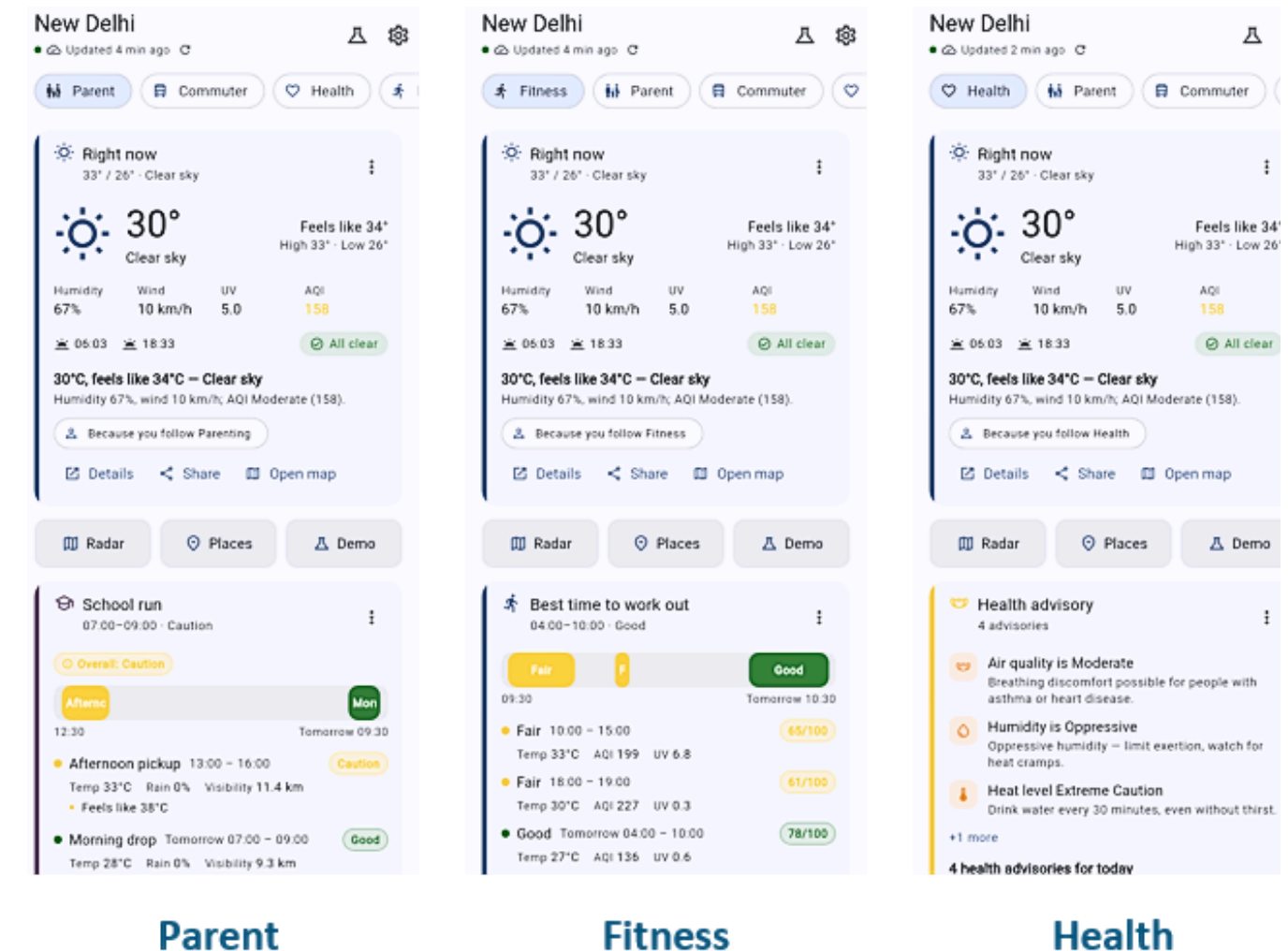
Lightweight, works offline, and stays accessible.

PROTOTYPE

pytest -q → 386 passed · flutter test → 125 passed
flutter analyze → clean · 10/10 demo steps PASS
63 screenshots · 115 commits

Flutter (Frontend): onboarding, ranked home, 15 renderer kinds across 33 card types, full-screen detail page each, radar map, saved places, settings, demo sheet, WebSocket client, offline cache, en/hi complete.

FastAPI (Backend): provider chain, 33 card builders, the engine, auth, saved places, events, admin console, /ws/alerts, Docker + Render config, GitHub Actions CI.



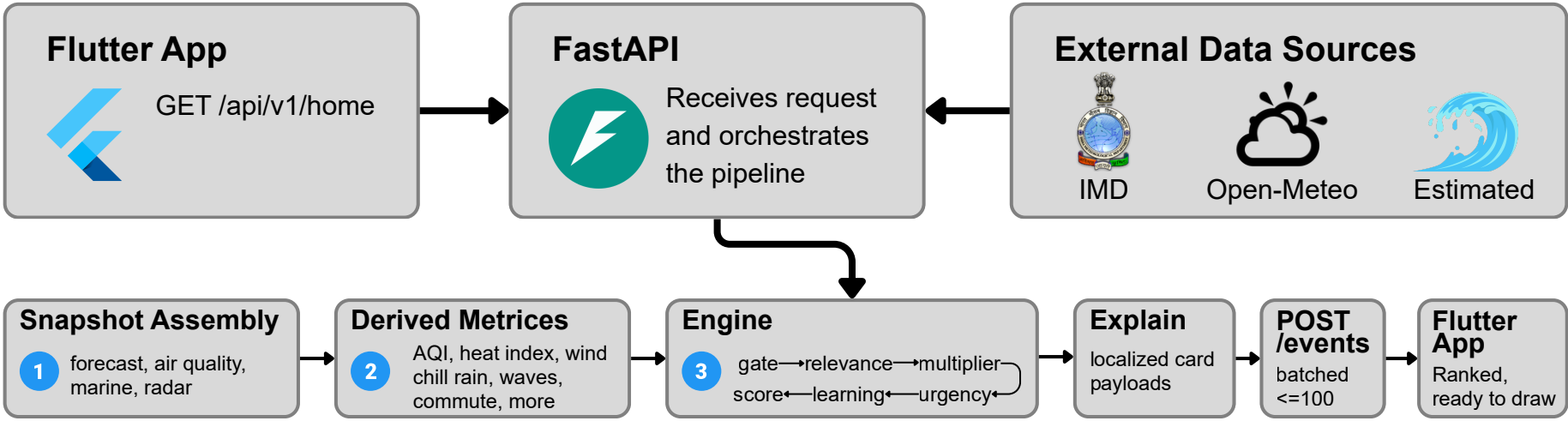
Parent

Fitness

Health

TECHNICAL APPROACH

ARCHITECTURE & RESILIENT DATA PIPELINE



Tech Stack

	MOBILE	Flutter 3.47 (Android/iOS/web) · Riverpod · go_router · Dio · flutter_map · fl_chart · ARB
	BACKEND	FastAPI · Python 3.13 · Pydantic v2 · httpx · SQLAlchemy 2
	ENGINE	Pure Python, no I/O — catalog · context · scoring · explain · learning · 33 builders
	STORAGE	SQLite default → PostgreSQL via DATABASE_URL
	CACHE	In-process TTL → Redis via REDIS_URL
	LIVE ALERTS	WebSocket /ws/alerts (FCM = production transport, designed)
	AUTH	Guest token + demo OTP, JWT HS256
	DEPLOY	Docker · infra/render.yaml · APK via GitHub Actions artifact

THE SCORING FORMULA

1 GATE
Eligibility

- Remove hidden cards
- Remove cards whose preconditions fail

Tide → Coastal location
Warnings → Active warning
Packing → Saved place
Heat → feels_like ≥ 35°C

2 RELEVANCE
User Fit
$$\text{rel} = \min(1, \max(\text{contribs}) + 0.15 \times (\sum \text{contribs} - \max(\text{contribs})))$$

Contribution =
Persona Weight x Card affinity

Primary persona: 1.0
Secondary personas: 0.7
Base relevance: 1.0

3 CONTEXT
When it Matters
$$\text{ctx} = \text{clamp}(\text{time} \times \text{season}, 0.4, 1.6)$$

- Time-of-day multiplier
- seasonal multiplier

Keeps context impact between 0.4 - 1.6

4 URGENCY
Data Driven

Derived directly from the current data:

Red heatwave : 1.0
AQI Very poor : 0.75
Dense fog : 0.8
Calm sea : 0

5 ENGAGEMENT
Learn From User
$$\text{eng} = 0.25 \times \tanh((\text{taps} + 2 \times \text{expands} + 3 \times \text{pins} - 3 \times \text{dismisses}) / 8)$$

- Taps → positive
- Expands → stronger positive
- Pins → strongest positive
- Dismisses → negative

Range: -0.25 to +0.25



FINAL SCORE

$$\text{Score} = 0.5 \times \text{rel} \times \text{ctx} + 0.5 \times \text{urg} + \text{eng}$$



DATA SOURCES

IMD current_wx_api · nowcastapi · warnings_district_api · aws_data_api

Status:
401 — needs IP whitelisting. Provider detects it, backs off 10 min, falls through

Used for:
station obs, 3-h nowcast, district warnings

Open-Meteo forecast / air / marine / geocoding

Status:
keyless 200

Used for:
current + hourly + 16-day, UV, visibility, soil moisture & temp, sunrise/sunset; PM2.5/PM10/O₃/NO₂/SO₂/CO → CPCB scale; waves, swell, SST

BigDataCloud reverse geocode · RainViewer

Status:
keyless 200

Used for:
lat/lon → district/state; radar frames + tiles

Open-Meteo forecast / air / marine / geocoding

Status:
optional free key

Used for:
station AQI / real traffic — estimated without them

FEASIBILITY AND VIABILITY

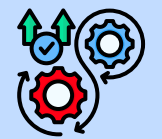
RIGOR, METRICS & RISK MITIGATION

FEASIBILITIES



TECHNICAL

Already green in CI. The engine is I/O-free, so scaling /home only scales a cache; the warm path stays in-process at ~2 ms.



OPERATIONAL

Zero-setup by design: no API key, Docker, or DB server. SQLite initializes on first boot, with Render deployment via blueprint. One script takes a fresh machine from clone to passing tests.



ECONOMIC

All default data sources are free and keyless. Paid sources fall back to labelled estimates, while the demo runs on free Render hosting.



INSTITUTIONAL

Server-driven cards turn homepage updates into backend deployments, not app-store releases. The IMD provider is ready and can be enabled by whitelisting—no code change needed.

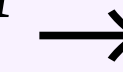
EVIDENCE

- Card types · renderer kinds · personas 33 · 15 · 8
- Payload 34.5 KB full → 13.5 KB lite
- Scenario overlays 10 / 10
- Warm /home 2 ms server-side · 4.6 ms median round trip · 3.6 ms under
- Release APK 62.5 MB, 3 ABIs, minSdk 24 / targetSdk 36
- Languages offered · complete 5 · 2
- Cities · coastal points in the offline gazetteer 212 · 102
- "Show less" taps to demote a card out of the feed 3 (rank 3 → 8, score 0.470 → 0.268)

RISKS → MITIGATION



IMD APIs need whitelisting, answer 401 from any un-approved host



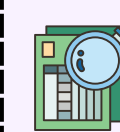
providers/imd.py handles 401s and retries after 10 min. Once approved, IMD goes live via whitelisting—no code changes.



A judged demo depends on the network



Offline demo: cached home renders with freshness; if cleared, "Sample data" appears.



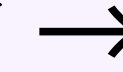
Modelled values mistaken for observations



"source": "estimated" + Estimated chip + tides disclaimer, enforced by the product principles in VISION.md and verified in QA step 4d.



A ranker that cannot be explained cannot be operated



≤4 reasons/card; formula in docs/03; 9 tests cover determinism, warnings, coastal gating, and personas.



Learning could bury a safety-critical card



Engagement stays within ±0.25; urgency ≥0.8 keeps red warnings pinned.

IMPACT AND BENEFITS

ECONOMIC BENEFITS

Protects farmer income by reducing fertilizer losses and improving efficiency, while mitigating weather-related economic risks for coastal communities.

ENVIRONMENTAL BENEFITS

Root-zone moisture, rainfall, AQI, Douglas Sea State, and harmonic tides help farmers, communities, and coastal users make safer environmental decisions.

SOCIAL BENIFITS

CPCB AQI categories prompt mask and asthma advisories, helping vulnerable groups take timely precautions against harmful pollution.

SAFETY

Orange+ pins warnings for every persona and pushes them to connected users in milliseconds. (17.7 ms measured)

TIME

The answer is on the first screen, not three taps down a menu of tables.

TRUST

Every card shows its reasons, provider status stays visible, and modelled data is marked "Estimated." Trust comes first.

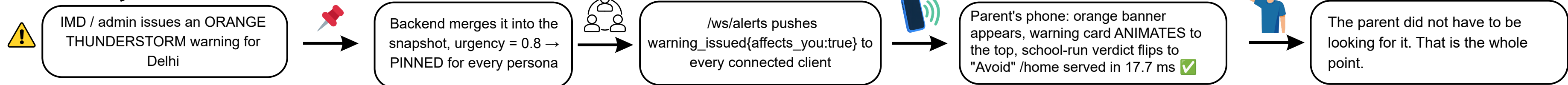
REACH

Hindi support, low-bandwidth mode, offline cache, Android 7.0+, TalkBack, and AA contrast—built for all devices, not just flagships.

OPERABILITY FOR IMD

Server-driven cards enable backend homepage control without release dependencies. IMD provider activates via whitelisting.

A LIVE JOURNEY



ALIGNMENT



UN "Early Warnings for All" (2027): addresses the last-mile delivery challenge



Digital India/Bhashini: enables regional-language delivery of government services



NATIONAL-SCALE IMPACT

A THE INCUMBENT'S OWN STORE PAGE

Mausam
★ 3.13/5
from 4.4k ratings

958M Indian internet users = ~0.1% reach

Top reviews complain about(exactly what this prototype fixes):

- No saved favourites
- District-level only
- Charts not understandable
- Slow

B THE HARM

Lightning
2,558 deaths(2023)

Fog road deaths
15,115 (2024)

Heat
733 (2024, independent count)

C THE MULLTIPLIER

24 h notice cuts damage 30%
(UN SG/ GCA)

1:9 benefit-cost

23,000 lives+
\$162B/yr
Globally available from better warnings

D OUR PROMISE

0.30 X 18,406
≈5,500 lives/yr
(upper bound)

(real world adoption, false alert. access, etc)
~50% (conservation)

"Every 1% converted in 55 lives a year."

E GROWTH

958M Online India

1.14B smartphones in India

~95% village 4G coverage

4X urban

So distribution is not the constraint; product experience is, and that is this PS

OUR WORK

Github repository

 github.com/Shreyansh303/team_mausam_sih_2026

API Contract

 REST + WebSocket normative for both sides [docs/04_api_contract.md](#)

QA Evidence

 10/10 demo steps [docs/QA_report.md](#)

Personalization engine spec

 Formulae, tests, explainability [docs/03_personalization_engine.md](#)

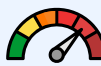
Card Catalog

 All 33 cards, affinities, gates, urgency rules [docs/02_card_catalog.md](#)

Push Design

 FCM as production alert transport [docs/09_push_notifications.md](#)

STANDARDS & SCIENCE WE IMPLEMENT

WHAT	STANDARD	WHERE USED	CODE/REF
 Air quality index	CPCB	National AQI, 6 bands,sub index per pollutant	services/aqi_cpcb.py
 Heat index	NWS Rothfusz	Regression + both adjustments	services/comfort.py
 Dew point	Magnus	formula	services/comfort.py
 Sea State	Douglas	scale	services/marine.py
 Wind	Beaufort	scale	services/marine.py
 UV Bands	WHO	Global Solar UV Index	builders/uv_index.py
 Warning colours	IMD	yellow / orange / red conventions	AppTheme.warningSeverityColor
 Rainfall intensity	IMD	classification	builders/rain_alert.py
 Agro-Zones	7 Indian Zones x 12 months		data/planting_calendar.json

DATA SOURCES

SOURCES	ROLE	ENDPOINT/NOTES
 IMD	Primary, whitelisting pending — station obs, nowcast, district warnings	mausam.imd.gov.in/api
 Open-Meteo	Fallback, keyless — the prototype runs on this today	api.open-meteo.com
 RainViewer	Radar frames and tiles	api.rainviewer.com
 BigDataCloud	Reverse geocoding	api.bigdatacloud.com
 CPCB · TomTom	Optional, keyed — station AQI, real traffic flow	api.bigdatacloud.com
 INCOIS / Survey of India	Named in the tides disclaimer as the authoritative source we are not using	

COMPETITIVE LANDSCAPE

PRODUCT	WHAT IT DOES	WHAT IT DOES NOT DO
 Mausam (IMD, today)	Authoritative Indian data, official warnings	One screen for everyone; no persona, no ranking, no explanation
 AccuWeather/Weather.com	Polished global apps, some personalization	Not IMD-authoritative for India; no CPCB AQI; no agro/tide layer for Indian users; closed ranking
 Windy/Ventusky	Excellent maps for enthusiasts	Expert-facing, not a citizen homepage; no warnings-first ranking
 Damini (lightning), Meghdoot (agri)	IMD's own targeted apps — right idea, right audience	Separate apps per audience. Our approach is the inverse: one app that becomes the right app.
Team Mausam	Persona x context x urgency ranking, explainable, server-driven, offline, bilingual	-

MARKET / ADOPTION CONTEXT

METRIC	VALUE/INSIGHT	SOURCE
 Weather-app market size / CAGR	~\$7.8B (2024) ~12.8% CAGR (2024-2030)	Grand View Research/MarketsandMarkets
 India smartphone	~1.2B (2024)	TRAI(subscription data)
 Rural internet users	~52%(2024)	TRAI(subscription data)
 Government digital-services adoption	~67%(2024)	Digital India / MeitY dashboards